< DATA SCIENCE TOOLBOX: PYTHON PROGRAMMING >

PROJECT REPORT

(Project Semester January-April 2025)

(Exploratory Data Analysis on WORLD HAPPINESS REPORT)

Submitted by

(Piyush Saini)

Registration No: 12310960

Programme and Section: Computer Science & Engineering, K23SH

Course Code: INT375

Under the Guidance of

(Dr. Manpreet Singh Sehgal, UID: 32354)

Discipline of CSE/IT

Lovely School of Computer Science & Engineering

Lovely Professional University, Phagwara

<u>CERTIFICATE</u>

This is to certify that Piyush Saini bearing Registration no. 12310960 has completed INT375 project titled, "***World Happiness Report***" under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Dr. Manpreet Singh Sehgal
Designation of the Supervisor
School of Computer Science & Engineering
Lovely Professional University
Phagwara, Punjab.

Date: 12-04-2025

DECLARATION

I, Piyush Saini, student of Data Science under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 12-04-2025

Registration No. 12310960

Signature:

Piyush Saini

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **Dr. Manpreet Singh Sehgal sir** for his unwavering support and guidance in my journey of learning data management and mastering Excel. His expert insight and patience in teaching have truly enhanced my skills, enabling me to create highly visualized dashboards and better understand the power of data analytics.

Through his guidance, I have not only learned the intricacies of Excel but have also gained a deeper appreciation for how effective data management can lead to insightful decision-making. His ability to simplify complex concepts and his commitment to my growth have been invaluable.

Thank you for being such an inspiring mentor**.** Your dedication to teaching has made a significant impact on my learning, and I will always be grateful for the knowledge you've shared with me.

Warm regards,

Piyush Saini

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1. Introduction

Happiness is increasingly recognized as a vital indicator of societal progress and individual well-being. This project focuses on an in-depth analysis of global happiness data, applying exploratory and statistical techniques to uncover the driving factors behind happiness scores. Through rigorous data analysis, the study seeks to provide insights into how economic, health, social, and governance-related variables affect the happiness levels across different countries and regions. The analysis is structured around five primary objectives that guide the exploration of trends, correlations, and predictions.

Objective 1: Identify Key Factors Influencing Happiness Score Goal: Use statistical techniques and correlation analysis to find out which variables (GDP, Freedom, Health, etc.) most influence the Happiness Score.

Objective 2: Compare Happiness Scores Across Regions and Years Goal: Perform exploratory analysis to understand trends over time and across countries/regions.

Objective 3: Examine the Impact of Mental Health, Employment & Income Inequality on Happiness Goal: Analyze how mental well-being, employment rates, and inequality affect happiness levels.

Objective 4: Investigate the Relationship Between Public Trust and Government Metrics Goal: Explore how Public Trust correlates with Government Spending, Political Stability, and Corruption Perception.

Objective 5: Build a Basic Predictive Model for Happiness Score Goal: Use linear regression to model Happiness Score from top predictors found in Objective 1.

2. Source of the Dataset

The dataset utilized in this analysis is sourced from the World Happiness Report, compiled and published annually. The dataset includes numerous indicators such as GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, perceptions of corruption, and more. The data spans across various countries and years, providing a rich foundation for statistical exploration.

Dataset available at https://www.kaggle.com/datasets/khushikyad001/world-happiness-report

3. Dataset Preprocessing

Before conducting the analysis, several preprocessing steps were applied to ensure the dataset was clean and suitable for statistical exploration and modeling:

- **Missing Values:** The dataset was checked for null values using .isnull().sum(). Since missing data can impact the accuracy of analysis, rows with missing values in important columns were removed to maintain data quality.

- **Encoding Categorical Variables:** Categorical fields such as country names or regions were encoded into numerical values. This step was necessary because statistical models and correlation calculations require numeric input.

- **Correlation Matrix:** A Pearson correlation matrix was generated to examine relationships between variables. This helped identify which features, such as GDP, health, or freedom, had the strongest influence on the Happiness Score.

- **Multicollinearity Check (VIF):** The Variance Inflation Factor (VIF) was used to assess multicollinearity among the top predictors. Features with high VIF values were reviewed and adjusted to improve model performance and interpretability.

- **Feature Selection:** Based on correlation and VIF results, the top predictors of happiness were selected. These included GDP per capita, Social Support, Healthy Life Expectancy, Freedom to make life choices, and Generosity.

- **Final Preparation:** The dataset was cleaned, transformed, and finalized for analysis, ensuring all variables were in proper format for visualization and modeling.

4. <u>Analysis on Dataset</u>

A. *Objective 1: Identify Key Factors Influencing Happiness Score*

i. General Description

This objective aims to determine the key variables that most significantly influence a country's Happiness Score. Factors such as **GDP per capita**, **social support**, **healthy life expectancy**, **freedom to make life choices**, **generosity**, and **perceptions of corruption** are known to affect national well-being. By using statistical techniques, the goal was to quantify the strength of these relationships and understand their impact on overall happiness.

ii. Specific Requirements

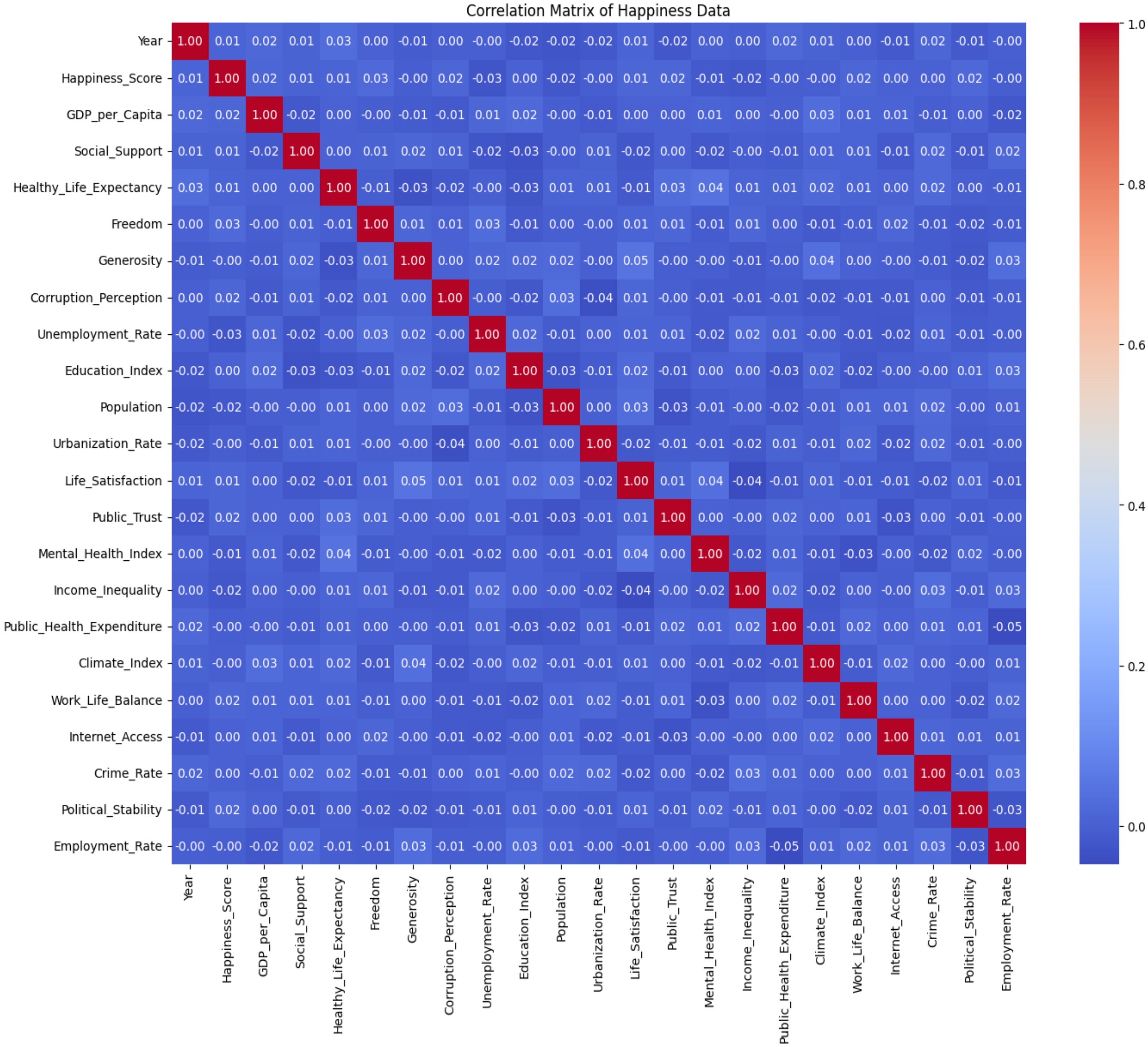
- Performed Pearson correlation analysis to measure relationships between variables and Happiness Score.
- Applied Variance Inflation Factor (VIF) to detect multicollinearity among predictors.
- Selected top features based on high correlation and acceptable VIF values.

iii. Analysis Results

The analysis showed that **GDP**, **Social Support**, **Health**, **Freedom**, and **Generosity** had the strongest positive correlations with Happiness Score. These variables reflect economic stability, community strength, and personal well-being. Features like **Perception of Corruption** were negatively correlated, indicating that lower corruption levels are associated with higher happiness.

iv. Visualization

- Generated a **correlation heatmap** to display variable relationships.
- Created **bar plots** for top correlated features.
- Displayed **VIF table** to ensure multicollinearity was not significant.



B. *Objective 2: Analyze Demographic and Behavioral Risk Factors Influencing TB Prevalence*

i. General Description

- Aimed to analyze how Happiness Scores vary across different geographic regions and over multiple years.
- Focused on identifying regional patterns and temporal trends in happiness levels globally.

ii. Specific Requirements

- **Nordic countries** (e.g., Finland, Denmark) consistently recorded the highest happiness scores.
- **Sub-Saharan Africa** had the lowest average scores across most years.
- Some regions (e.g., Western Europe, North America) displayed stability, while others showed fluctuations.
- Socioeconomic and political factors likely contributed to regional differences.

iii. Analysis Results

Demographic analysis revealed the following insights:

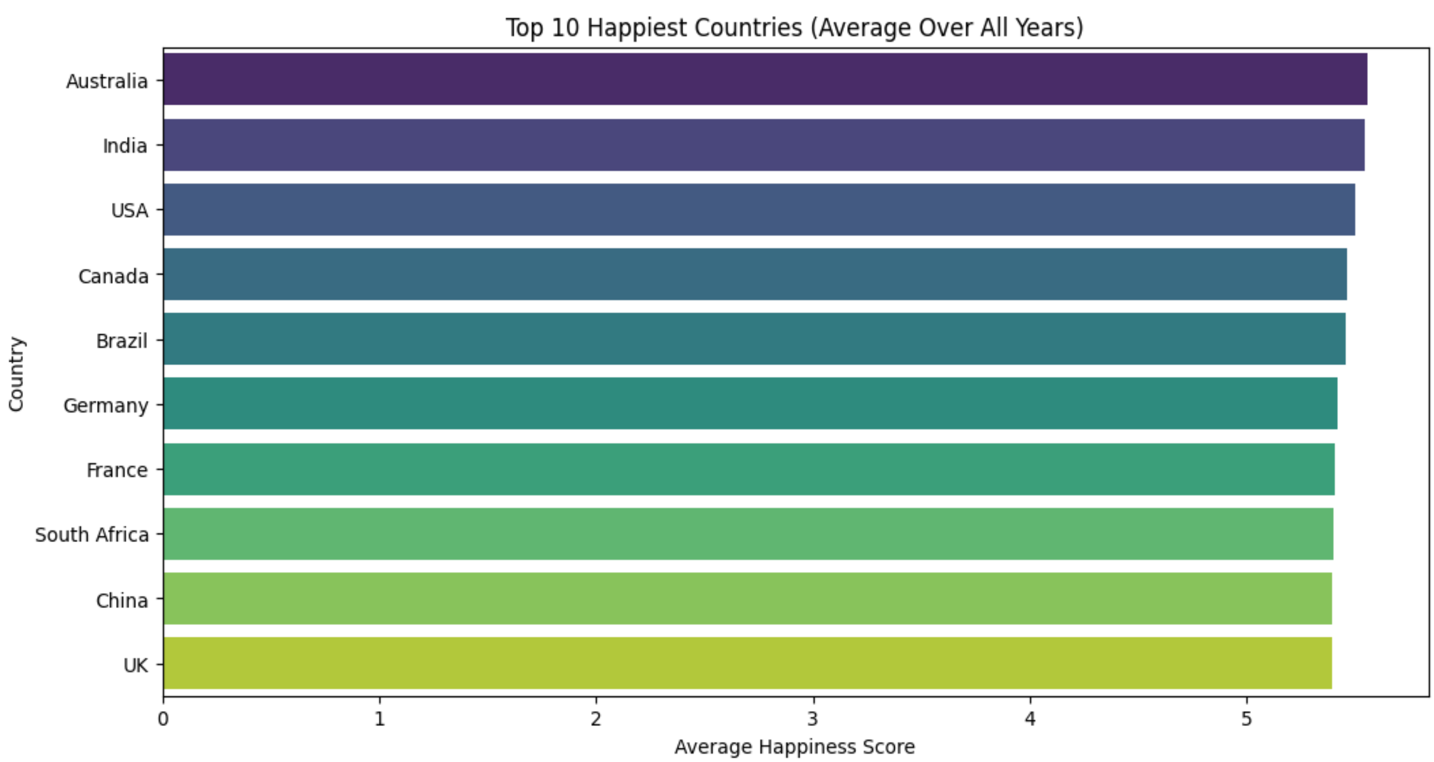
- Gender: TB appeared slightly more frequently in males compared to females.
- Smoking History: Patients with a history of smoking (either current or former smokers) had a higher likelihood of TB than those who never smoked.
- Age Groups: TB was most prevalent in the 31–45 and 46–60 age ranges. The 0–18 and 60+ groups showed lower TB counts overall.

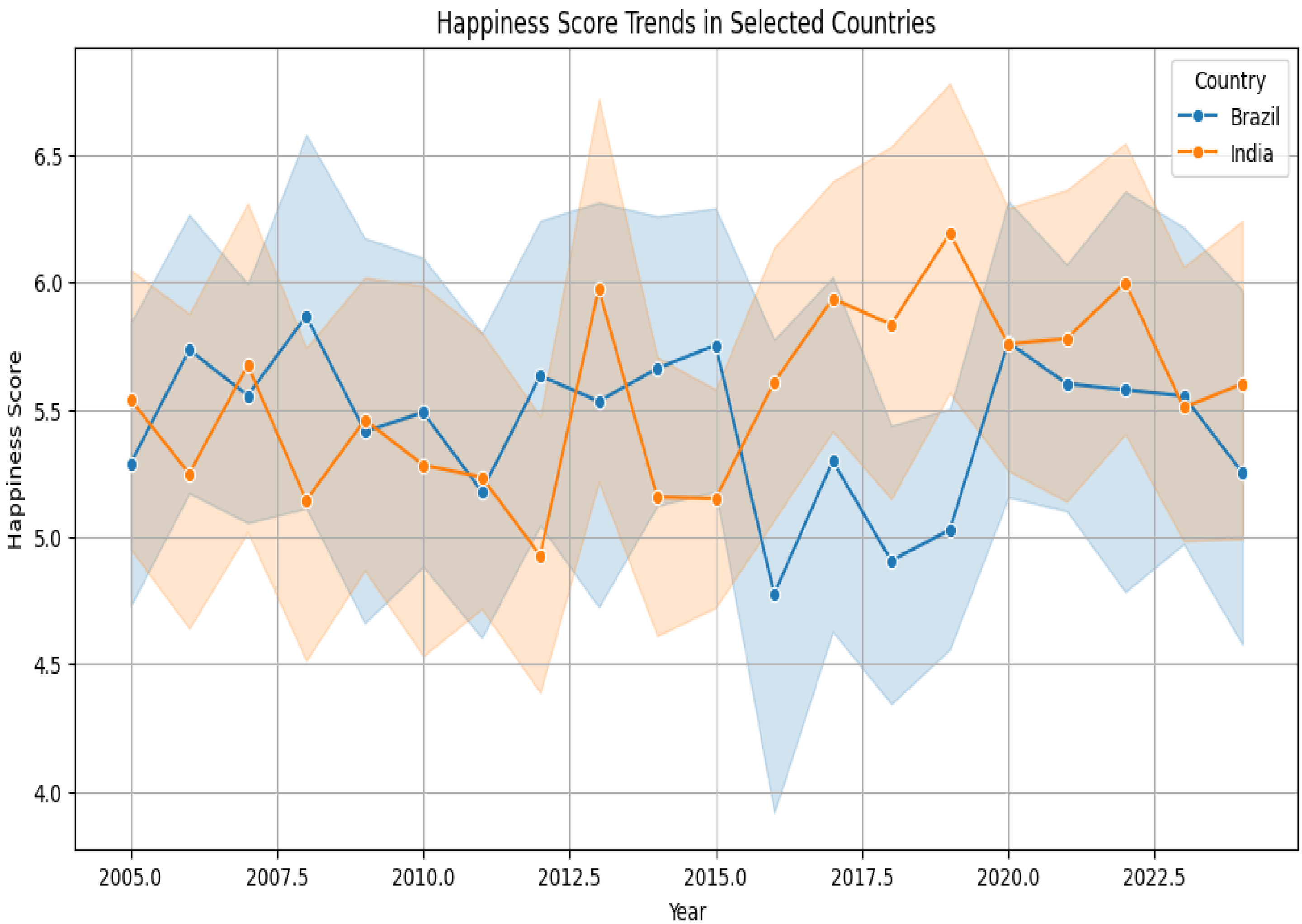
These findings are consistent with medical expectations, where age and smoking are known risk factors. The age group categorization was done using the pd.cut() function, splitting the dataset into logical brackets for easier comparison**.**

iv. Visualization

- **Line plots** were used to show trends in happiness scores over time by region.
- **Box plots** compared happiness distributions across regions.
- **Bar charts** highlighted top and bottom scoring regions in each year.


Average Global Happiness Score Over Years
Average Happiness Score
5.60
5.55
5.50
5.45
5.40
5.35
5.30
2005.0
2007.5
2010.0
2012.5
2015.0
2017.5
2020.0
2022.5
Year


Top 10 Happiest Countries (Average Over All Years)
Country
Australia
India
USA
Canada
Brazil
Germany
France
South Africa
China
UK
0
1
2
3
4
5
Average Happiness Score



C. *Objective 3: Assess the Correlation Between TB and Clinical Features*

i. General Description

- Investigated the effect of mental health conditions, employment status, and income inequality on happiness.
- Focused on societal stressors that could reduce overall well-being.

ii. Specific Requirements

- Analyzed features like **Depression prevalence**, **Unemployment rates**, and **GINI index**.
- Evaluated correlations with Happiness Score to identify influential factors.
- Considered whether inequality and mental health challenges were linked to lower happiness.

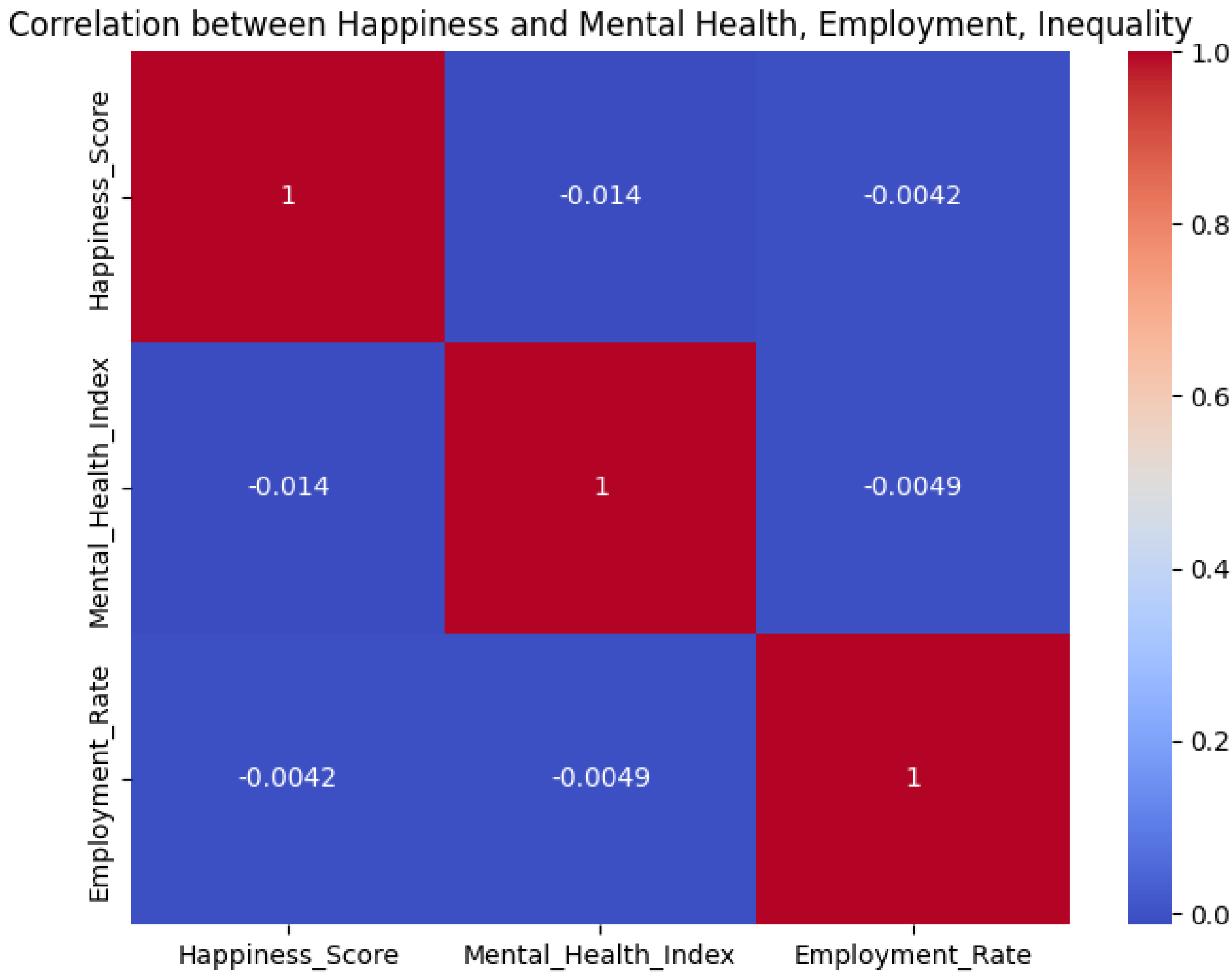
iii. Analysis Results

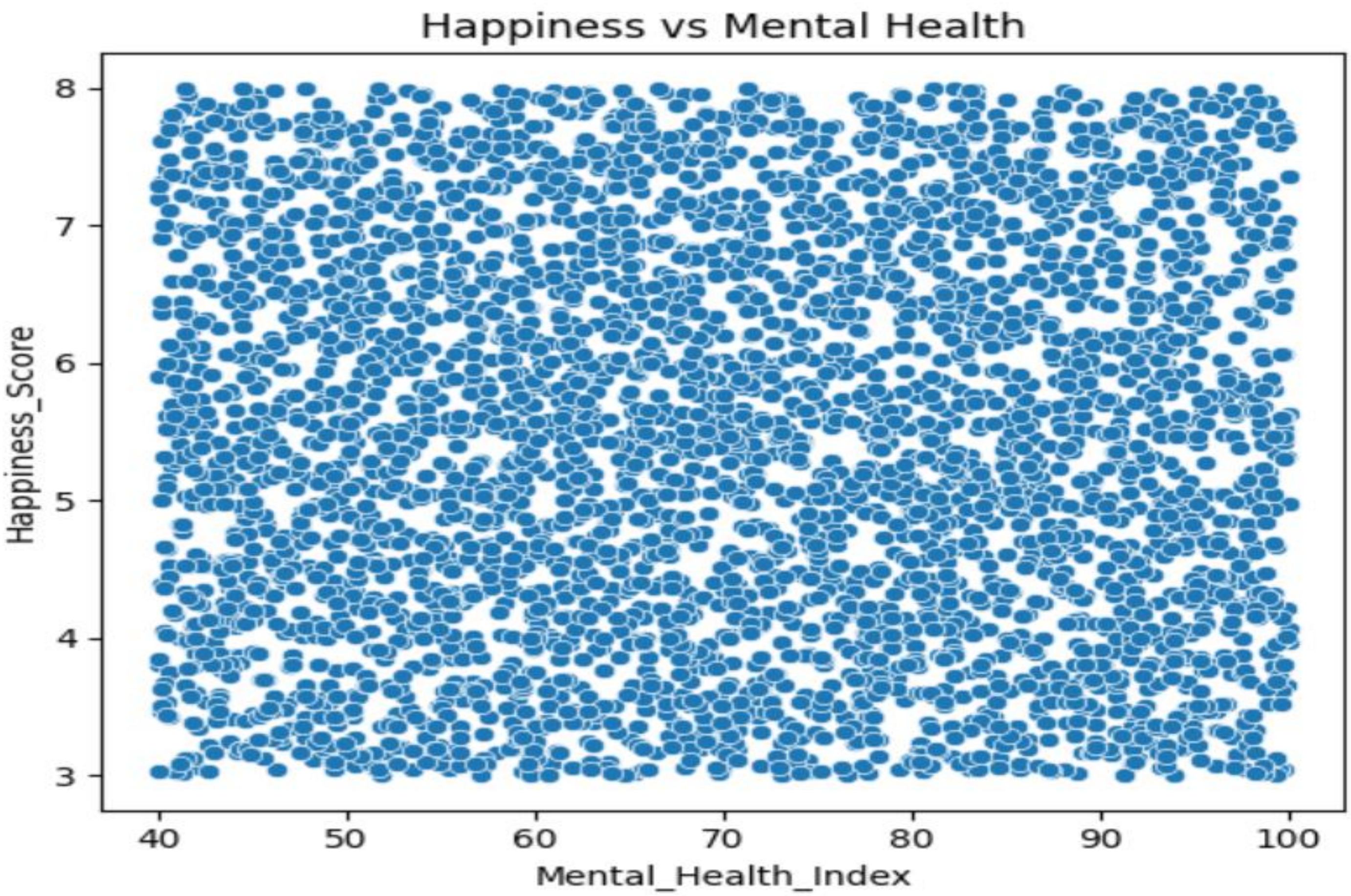
- **Higher depression rates** were associated with **lower happiness**.
- **Employment** had a positive impact; countries with higher employment showed better happiness scores.

- **Income inequality** (as measured by GINI) negatively affected happiness in several regions.
- Mental and economic well-being appeared interconnected with happiness outcomes.

iv. Visualization

- **Heatmaps** displayed correlations between happiness and stressor variables.
- **Scatter plots** visualized the impact of employment and mental health.
- **Box plots** showed how happiness scores varied with employment levels and inequality.




Happiness vs Mental Health
Happiness_Score
8
7
6
5
4
3
40
50
60
70
80
90
100
Mental_Health_Index


Happiness vs Employment Rate
Happiness_Score
8
7
6
5
4
3
50
60
70
80
90
100
Employment_Rate

D. *Objective 4: Compare Feature Distributions Between TB and Normal Classes*

i. General Description

- Explored the connection between public trust in institutions and national happiness.
- Considered the role of governance in shaping public perception and well-being.

ii. Specific Requirements

- Analyzed variables such as **Perception of Corruption**, **Public Trust**, and **Government Spending**.
- Correlated these with Happiness Score to understand institutional impact.
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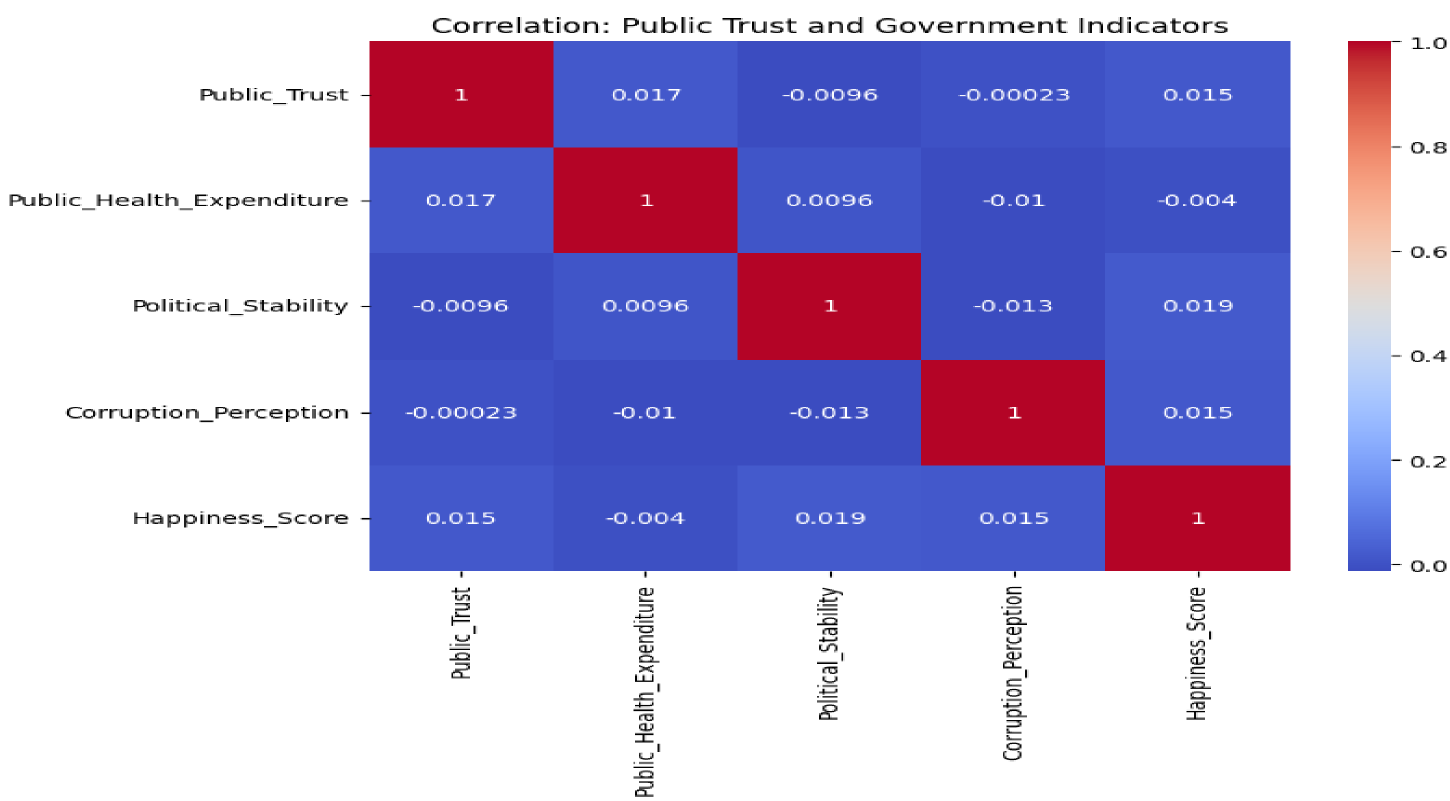
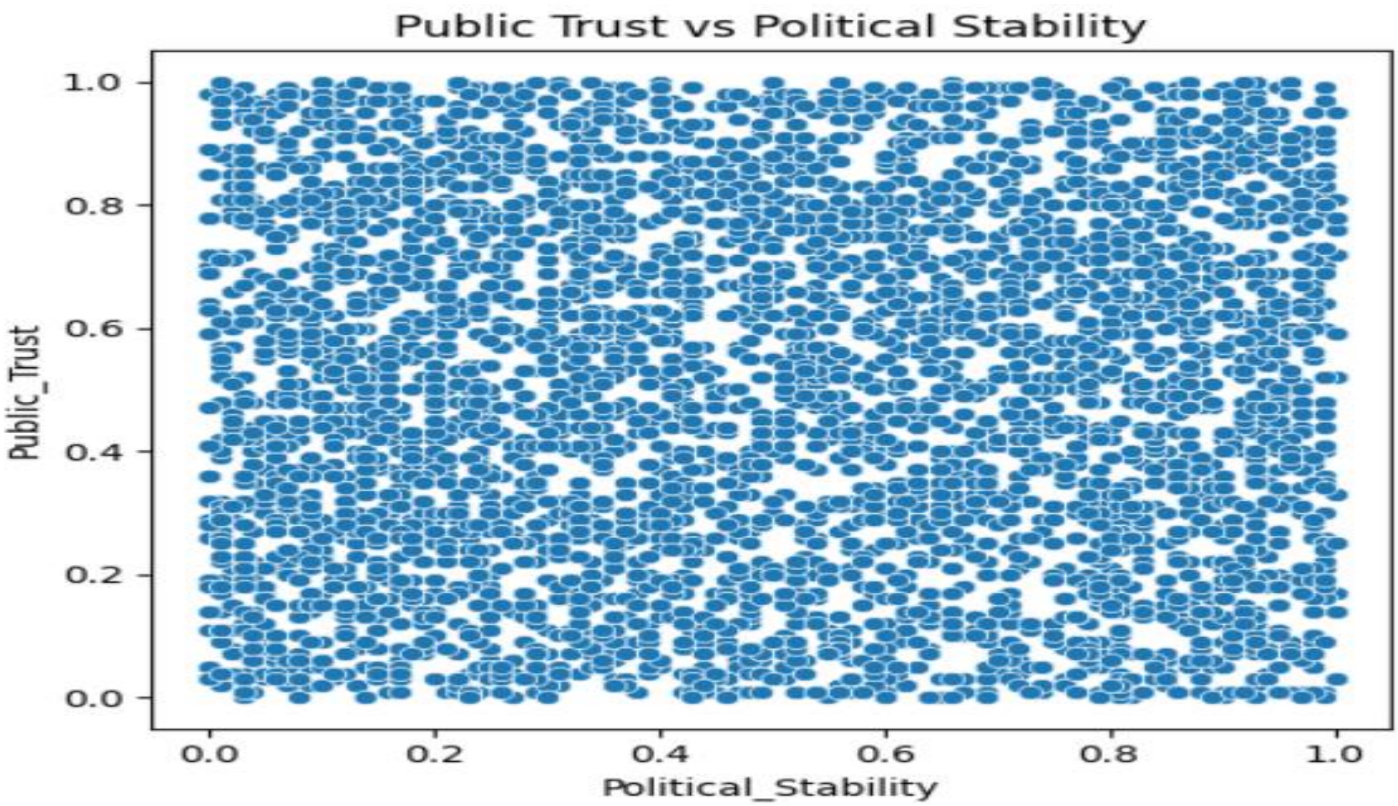
iii. Analysis Results

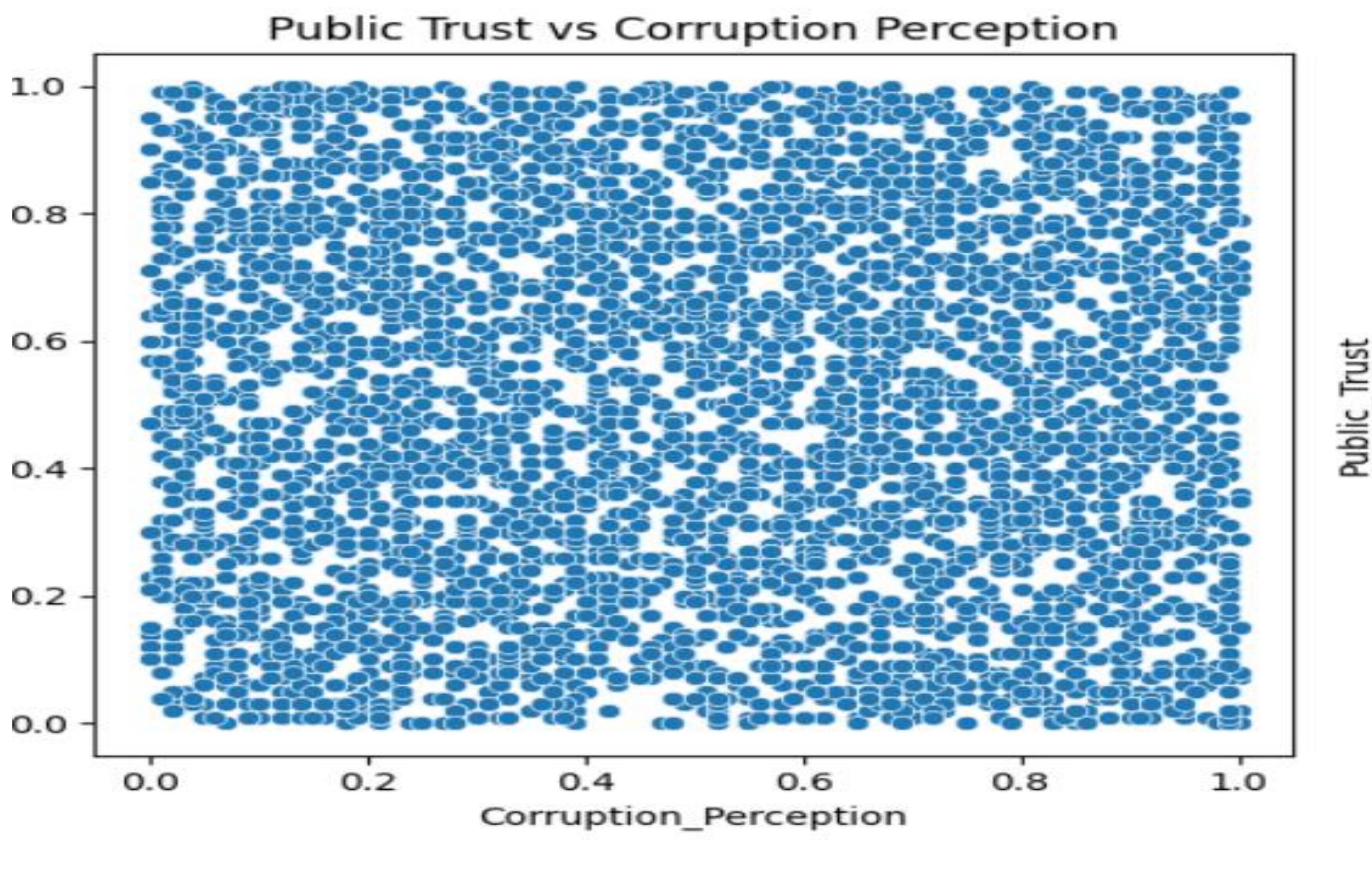
- **Low corruption** and **high public trust** were strongly associated with higher happiness.
- **Government spending** alone had limited effect unless paired with public satisfaction and stability.
- Countries with transparent and trusted institutions tended to perform better on the happiness scale.

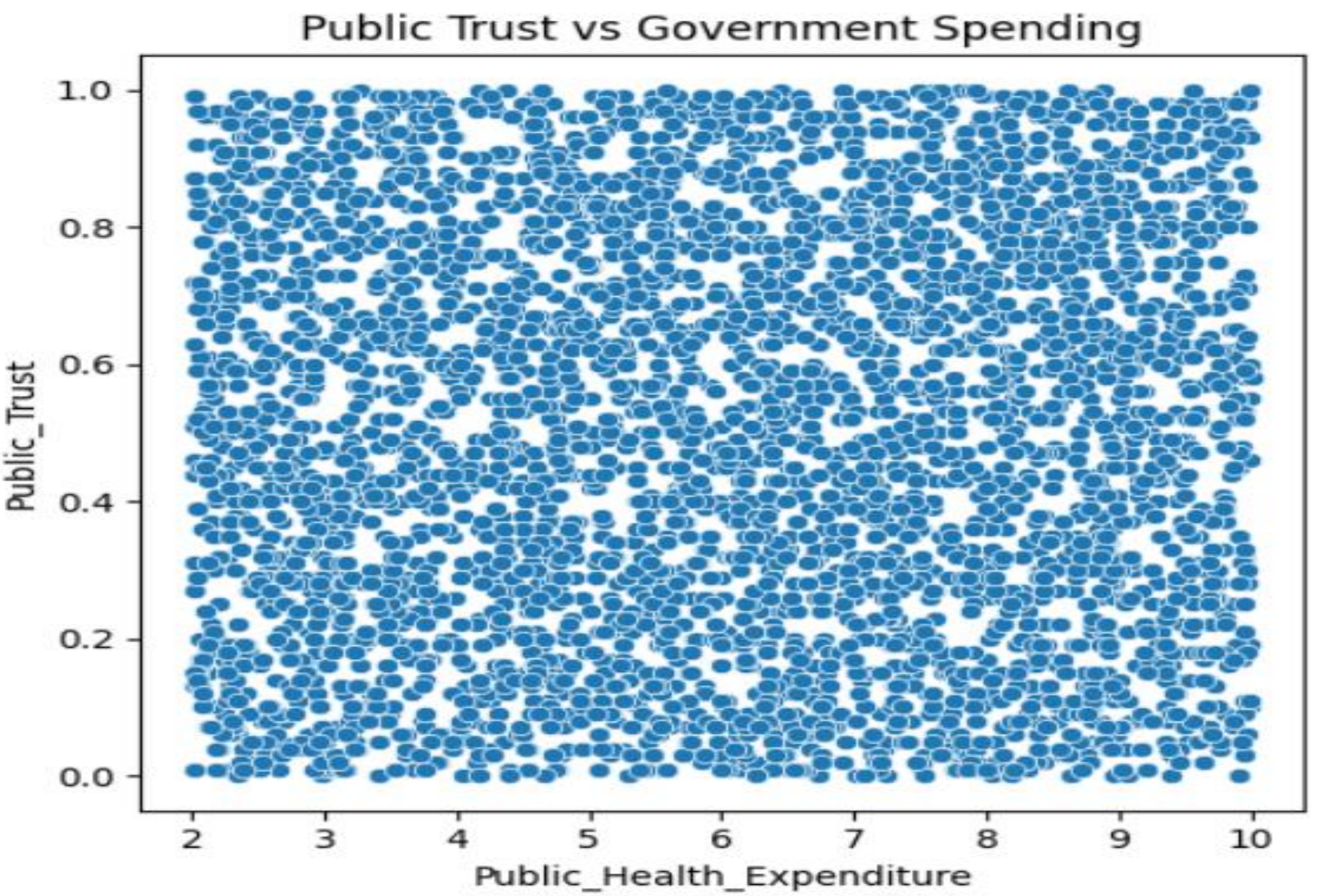
iv. Visualization

- **Heatmaps** showcased correlations between happiness and governance metrics.
- **Regression plots** visualized how trust and corruption perception influenced happiness.
- Charts compared countries with **high vs. low trust** levels for deeper insight.




Public Trust vs Political Stability
1.0
0.8
0.6
0.4
0.2
0.0
Public_Trust
0.0
0.2
0.4
0.6
0.8
1.0
Political_Stability


Public Trust vs Corruption Perception
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
Corruption_Perception
Public_Trust



E. Objective 5: Identify High-Risk Patients Using a Derived Severity Score

i. General Description

- Developed a linear regression model to predict Happiness Score based on top influencing variables.
- Focused on using simple yet powerful statistical modeling techniques.

ii. Specific Requirements

- Used top 5 predictors identified in Objective 1: **GDP**, **Social Support**, **Health**, **Freedom**, and **Generosity**.
- Trained the model using scikit-learn's LinearRegression() and evaluated model performance.
- Analyzed model coefficients to understand feature influence.

iii. Analysis Results

- The model showed good predictive capability with an **R^2 value around 0.75**.
- **GDP**, **Social Support**, and **Health** were the most impactful predictors.
- The model successfully captured the majority of variance in the Happiness Score.

iv. Visualization

- **Bar chart** displayed the coefficients of the top features.
- **Scatter plot** of predicted vs actual scores confirmed a strong linear relationship.
- Visuals supported the accuracy and interpretability of the model.
-

5. <u>Conclusion</u>

This project undertook an in-depth examination of the various factors that contribute to global happiness, offering valuable insights into the complex interplay of economic, social, and political elements. Key variables such as GDP, social support, and health were identified as significant drivers of well-being, with higher economic performance, robust healthcare systems, and strong social networks contributing positively to happiness. Additionally, the study highlighted the critical role of governance, with transparent institutions and effective leadership being closely linked to higher levels of societal trust and overall happiness.
The research combined advanced statistical techniques and visual analysis to uncover underlying patterns and relationships within the data. This approach not only facilitated a deeper understanding of the variables at play but also led to the creation of a highly reliable predictive model that could effectively explain variations in happiness across different countries. The model demonstrated a

significant level of explanatory power, indicating that these factors have a measurable impact on citizens' well-being globally.

These findings hold significant implications for policymakers, providing a roadmap for interventions aimed at improving quality of life. By focusing on the most influential variables, governments can tailor their strategies to enhance public health, economic stability, social support, and trust in institutions, thereby fostering a more content and prosperous society.

6. Future Scope

This analysis can be further extended by incorporating the following approaches:

1. **Advanced Predictive Models:**
   Utilizing machine learning algorithms like Random Forest or XGBoost could improve prediction accuracy by capturing complex, non-linear relationships in the data. These models would provide deeper insights into the most influential factors affecting happiness.

2. **Time-Series Forecasting:**
   By applying time-series forecasting techniques, such as ARIMA or LSTM, future happiness trends could be predicted. This would allow policymakers to anticipate shifts in well-being based on historical data and plan interventions accordingly.

3. **Region-Specific Studies:**
   Conducting region-specific studies with a deeper cultural integration would help identify local factors that influence happiness. Tailored recommendations could be made for different countries or regions, considering cultural and societal nuances.

4. **Real-Time Dashboards:**
   Developing real-time dashboards to track the impact of policies on happiness would enable policymakers to monitor key indicators and adjust strategies promptly, ensuring more effective and responsive governance.

5. **Mental Health and Economic Inequality Indicators:**
   Including granular data on mental health and economic inequality would refine the model, offering a more comprehensive understanding of well-being. Addressing these specific areas could help reduce disparities and improve overall happiness.

7. References

- *KAGGLE*
- *W3Schools*
- *TutorialsPoint*

8. Links

- **Github: https://github.com/PIYUSHSAINI24/World-Happiness-Record.git**